

Sentinel DV — AI-Powered MCP Server for Design Verification Intelligence

AI / MCP Clients

Claude / GPT

VS Code Copilot

Custom AI Agent

CI/CD Pipeline Bot

Model Context Protocol (MCP) — JSON-RPC 2.0 over stdio/SSE

25 MCP Tools (6 Categories)

Runs
runs.list · runs.get
runs.diff · runs.submit

Tests
tests.list · tests.get
topology · cluster

Coverage
list · summary · trend
gaps · advisor

Assertions
list · get · failures
sva_status · vacuity

Failures
failures.list
regressions.summary
health

Live Sim
wave.signals
wave.summary
sim.status

Sentinel DV Core Engine — DuckDB Store · Indexer · Normalization · Coverage Advisor · Redaction

Adapters (Parsers)

UVM Log Parser
VCS / Questa / Xcelium
UVM_ERROR · Test status

Coverage Parsers
JSON · XML · TXT
dashboard.html · overalldu.js

Assertion Reports
JSON · SVA status
Vacuity detection

Waveforms
VCD parser
Signal queries

JUnit XML
xUnit format
CI integration

Simulation Artifact Sources

VCS
URG coverage
UVM logs
Assertion JSON

Questa / ModelSim
vcover HTML
UVM logs
VRMDB

Xcelium / Cadence
IMC coverage
UVM logs

Verilator + CoCoTB
VCD waveforms
JUnit XML

CI / Jenkins
Artifact archives
Regression runs

Validated: 1,477 CI runs · 6 production projects · 3 simulator vendors · 4,814 real failures · 6 coverage reports · 187 unit tests

github.com/kiranreddi/sentinel-dv · Open Source (MIT)